

The Substrate Closure Principle

Casey Koons (named principle) + Keeper (formalization)

2026-05-20 Wednesday EOD

The Substrate Closure Principle

Casey-named statement (hypothesis register, per Cal Calibration #13)

“The BST framework operates AS IF the substrate is operationally closed. No substrate-external referent is identified in current observations — no simulator, no observer-outside-the-graph, no higher-level computational layer. Within the framework, the substrate appears to compute its own state by algebraic identity; substrate appears to BE both the computing AND the computed. Questions seeking substrate-external grounds are mal-posed within the framework as currently observed.”

This is a WORKING HYPOTHESIS, not a derived claim. The “as if” register is load-bearing: BST does not claim to have proven substrate’s metaphysical closure; it observes that no substrate-external referent has been required for any BST observable to date.

What the framework observes

In the AC theorem graph (currently 2155 nodes, 9769 edges), every reference is internal. No theorem requires substrate-external grounds. No observable points outside the substrate’s computational domain. The graph is closed and self-referential at current observational state.

This is structurally consistent with two competing ontologies: - Eternal substrate: no beginning required; computation is steady-state - Bootstrap from minimum: substrate self-bootstraps from minimal initial condition

The framework doesn’t distinguish (a) from (b) at current resolution.

What the framework does NOT claim

- NOT a simulation hypothesis refutation at metaphysical level: substrate-closure as observed cannot rule out a substrate-external observer in principle;

it only observes that BST observables don't require one

- NOT a “we have solved why there is something rather than nothing” claim: that question may be substrate-external and outside the framework's domain
- NOT closure under Tegmark-MUH at full modal-realism: BST narrows to D_{IV}^5 specifically; closure here is observational

Falsifier specification

The principle is falsified by discovery of any observation that requires substrate-external information to explain. Specifically: - Discovery of a physical phenomenon that cannot be derived from D_{IV}^5 + BST primaries + π (T719 Observable Closure boundary) - Discovery of computational signatures that imply external simulator structure (e.g., precision artifacts at “simulation resolution” boundaries) - Discovery of substrate-external information leaking into substrate observables (e.g., correlations with apparently-uncorrelated external structures)

Any single such observation invalidates substrate-closure within the framework.

Tier discipline (per Cal Substrate-Cognition External Register methodology)

Level 1 (D-tier supportable internally): - AC graph closure at current observational state (no substrate-external references) - T719 Observable Closure (BST observables in $\bar{Q}(\text{BST primaries})[\pi]$) - Six-audit cascade-unblock pathway operating substrate-internally

Level 2 (I-tier working hypothesis with falsifier): - The framework's substrate-closure is a structural property of substrate itself (not just current observations) - “Substrate IS the existing thing” ontological reading - Bostrom-style simulation hypothesis is mal-posed within framework

External register discipline (per Cal default-deny standard)

Internal use (this document + audit chain + team discussion): substrate-closure framing acceptable as working hypothesis.

External use (papers, outreach letters, Zenodo materials): DEFAULT-DENY UNTIL CAL GRADE-PASS.

Acceptable external register (operational, falsifier-bearing): - “BST framework requires no substrate-external degrees of freedom for X observables” - “BST observables are closed under $\bar{Q}(\text{BST primaries})[\pi]$ at current observational scope”

Unacceptable external register (triggers crank-pattern dismissal): - “Substrate IS reality” / “Substrate computes itself, no simulator required” - “BST refutes the simulation hypothesis”

Connection to other principles

- SWPP (Substrate Working Process Principle): SWPP describes WHAT substrate does (active information cycle); Closure describes WHAT IS substrate's domain (operationally closed at current observation)
- Integer Web Principle: integers as substrate boundaries holding webs; Closure addresses the OUTER boundary of substrate's domain
- Five-Absence: substrate doesn't contain GUT, proton decay, DM particle, etc.; Closure refines this with "substrate's domain has no external referent"

Per Casey's standard

- Simple: substrate's operations don't require external grounds (current observation)
- Works: every BST observable to date is substrate-internal
- Hard to break: would require finding substrate-external information leaking into substrate observables
- Counter-example: none currently observed

Filing status

Casey-named principle FILED per Wednesday 2026-05-20 approve-all directive. L2 working-hypothesis tier with explicit falsifier. Internal register acceptable; external default-deny until Cal grade-pass per substrate-cognition external register methodology.

— Keeper, 2026-05-20 Wednesday EOD (Cal register discipline applied)